Murat Ambarkutuk

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Professional Experience

R^2 Robotics - RANPAK **November 2024 – Present**
Principal Perception Engineer *January 2025 – Present*
Sr. Perception Engineer *November 2024 – January 2025*

- Architect the second-generation Decision Tower perception stack (full rewrite) for high-throughput packaging automation: multi-camera calibration and 3D reconstruction, YOLO-based real-time detection, TensorRT-accelerated inference, and neural scene representations.
- Built a production TSDF fusion pipeline with voxel hashing in CUDA for large-scale 3D mapping (depth-only and depth+color, GPU-optimized memory access), together with a Depth Regularizer for winsorized multi-view depth fusion with polynomial-regression weighting and robust confidence filtering.
- Built libinference, a header-only, cross-platform ONNX/TensorRT inference library with dynamic shapes, multi-GPU support, CUDA stream parallelism, and unified logging/configuration for seamless deployment across development and production environments.
- Maintain the release and dev channels of a private, S3-backed APT repository used to distribute software packages.
- Led model optimization: distilled DINOv3 features into lightweight YOLOv8 models for real-time, high-accuracy multi-class segmentation on embedded hardware; built a few-shot logo classifier (prototypical network on a frozen DINOv2 backbone) for brand and logo recognition.
- Pretrained a shared backbone on a standard architecture and defined the downstream contracts that let independent task heads solve many perception problems from one backbone.
- Fine-tuned and deployed a multi-head classifier on this backbone; amortizing inference across heads enabled edge-level optimizations for industrial-scale perception.
- Built synthetic data generation pipelines with NVIDIA Isaac Sim and a Blender-based native generator to produce simulated data for the perception models.
- Developed Quality Tower, a multi-class segmentation pipeline for on-line carton graphic validation and hazard-label inspection.
- Host cross-functional engineering meetings to align perception objectives with the product roadmap, and mentor perception team members on computer vision, deep learning optimization, and production deployment.

Baykar Defense (TR) - Atlas Imaging (US) **2018 – 2020**
Computer Vision Engineer *Remote*

- Developed orthorectified maps from aerial photos taken by autonomous UAVs. Derived the pose (location and orientation) of the UAV and its components from distributed Inertial Navigation System measurements, then used it in an iterative computer vision algorithm that projects image corners onto the ground while accounting for elevation changes in the scene.
- Developed a sensor fusion algorithm, with a statistical representation, to correct the results of the earlier stages of the map-making pipeline.
- Introduced test-driven development to the team: wrote tests for coverage, style requirements, and the performance and accuracy of the algorithms.
- Architected the CI/CD pipeline that runs these tests automatically and generates reports from the results.

Education

PhD in Computer Engineering **2018 – 2024**
The Bradley Department of Electrical and Computer Engineering at Virginia Tech *Blacksburg, VA*

- Paul E. Torgersen Research Excellence Award
- The Outstanding Research Award and Fellowship from the Turkish Ministry of National Education

Dissertation: A Sensor Fusion Technique for Gait Analysis from Structural Vibration and Computer Vision

MS in Mechanical Engineering **2015 – 2018**
Mechanical Engineering at Virginia Tech *Blacksburg, VA*
Thesis: A Grid based Indoor Radiolocation Technique Based on Spatially Coherent Path Loss Model

BS in Mechatronics Engineering **2008 – 2013**
Mechatronics Engineering Department at Kocaeli University *Kocaeli, Turkey*
Thesis: Box Dimensioning System with Kinect Sensor

Technical & Language Skills

Languages: Python, C++, C, CUDA, MATLAB
ML/CV Frameworks: PyTorch, TensorFlow, Keras, OpenCV, scikit-learn, scikit-image, YOLO, DINOv2/v3
3D Vision & Robotics: COLMAP, Multi-view Geometry, TSDF Fusion, 3D Gaussian Splatting, ROS
Simulation & NVIDIA Stack: Isaac Sim (Omniverse), Blender, Synthetic Data Generation, CUDA, TensorRT
Deployment, Optimization & MLOps: ONNX, MLflow, CUDA Kernels, Quantization, Model Distillation
Tools & Technologies: Linux, APT, AWS S3, Git, GitHub, GitLab, CI/CD, Docker, LaTeX
Language Proficiency: Turkish (Native), English (Proficient), German (Novice)

Research Projects

Sensor Fusion for Occupant Localization in Smart Buildings

Advisors: P Plassmann, CF Jones | Related publications: [1, 2, 5–7] **2018–2024**

- Developed a robust algorithm that estimates heel-strike locations from the accelerometer measurements of a vibrating floor.
- Built a computer vision algorithm that localizes heel-strike locations from stereoscopic camera images.
- Designed a sensor fusion framework that combines the vibration- and vision-based heel-strike locations.
- Created a combined simulation environment that generates the vibration and visual data needed to validate the algorithms.
- Quantified the efficacy and accuracy of the implementations with simulations and controlled experiments.

Quantification of Collaboration and Communication among Students in a Smart Classroom

Advisors: T Baird, PA Tarazaga | News Article **2018**

- Designed a multiple-view, camera-based sensor suite (frontend + backend) that quantifies collaboration among students in a smart classroom. The backend controls the sensors, schedules automatic start/stop times, and applies automated computer vision techniques to the collected data; the frontend exposes remote control and monitoring of the system.

A Grid-based Indoor Radiolocation Technique based on Spatially Coherent Path Loss Model

Advisor: T Furukawa | Publications: [4] **2016–2017**

- Derived a probabilistic framework for passive radio localization under multipath propagation that determines the location of occupants, robots, and inventory in indoor environments.
- Developed the localization algorithm based on the probabilistic framework.
- Validated the algorithm with controlled experiments whose results were presented at an IEEE RAS conference.

Relevant Publications

[1] S. Alajlouni, **Ambarkutuk, Murat**, and P. Tarazaga. Towards the detection and localization of multiple occupant footsteps from vibroacoustic measurements. In *Dynamics of Civil Structures, Volume 2: Proceedings of the 38th IMAC, A Conference and Exposition on Structural Dynamics 2020*, page 193. Springer, 2020.

[2] S. Alajlouni, V. V. S. Malladi, and **Ambarkutuk, Murat**. Investigation of using log-spectrum averaging (cepstral averaging) for blind reconstruction of an unknown impact input force. In *Dynamics of Civil Structures, Volume 2: Proceedings of the 39th IMAC*. Springer, 2022.

[3] H. Ocak, **Ambarkütük, Murat**, G. Küçükyıldız, and S. Karakaya. Image processing based package volume detection with kinect. In *2015 23nd Signal Processing and Communications Applications Conference (SIU)*, pages 515–518. IEEE, 2015.

[4] **Ambarkutuk, Murat** and T. Furukawa. A grid-based indoor radiolocation technique based on spatially coherent path loss model. In *2017 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)*, pages 220–226. IEEE, 2017.

[5] **Ambarkutuk, Murat** and P. E. Plassmann. Modeling and analysis of dispersive propagation of structural waves for vibro-localization. *Sensors*, 24(23):7744, 2024. doi: 10.3390/s24237744. URL https://doi.org/10.3390/s24237744.

[6] **Ambarkutuk, Murat** and P. Tarazaga. Uncertainty analysis of an occupant localization technique based on simulated structural vibrations. In *Dynamics of Civil Structures, Volume 2: Proceedings of the 39th IMAC*. Springer, 2022.

[7] **Ambarkutuk, Murat**, S. Alajlouni, P. A. Tarazaga, and P. E. Plassmann. A multi-sensor stochastic energy-based vibro-localization technique with byzantine sensor elimination. *Sensors*, 23(23), 2023. ISSN 1424-8220. doi: 10.3390/s23239309. URL https://www.mdpi.com/1424-8220/23/23/9309.

Community Service & Leadership

Graduate Student Ambassador, ECE at Virginia Tech · Student Member & Reviewer, IEEE Robotics and Automation Society · Student Member, Society of Experimental Mechanics · Secretary, Turkish Student Association at Virginia Tech